SMART INDIA HACKATHON 2026

TITLE PAGE

- **Problem Statement ID:** SIH 26263
- **Problem Statement Title:** Development of an APAAR ID-Based Student Data Management and Verification Platform for AICTE-Approved Institutions
- **Theme:** Smart Automation
- **PS Category:** Software
- **Team ID:** 199751
- **Team Name:** Phoenix Crew

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WAYMARK APAAR HUB

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PROBLEM

- PGDM institutions send AICTE detailed student data once a year. Everyone else gives a total enrolment figure at approval time.
- With only totals, AICTE can't tell who is actually enrolled, or whether one student sits in two colleges.
- Admissions above the approved intake turn up late, if they turn up.
- An employer or regulator who wants to check a student record has no easy way to do it.

OUR APPROACH

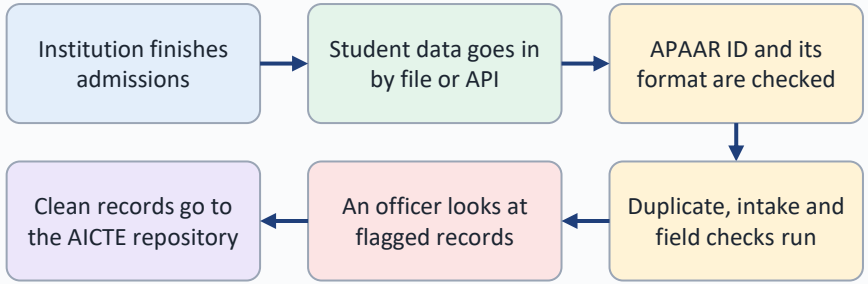
- Each student record hangs off one APAAR ID.
- Once admissions close, institutions upload by file or through an API.
- Each upload is checked on arrival for bad IDs and repeat students, and enrolment is compared with approved intake.
- AICTE watches the numbers live. Others can verify a record only if the student agrees.

WHAT IS DIFFERENT

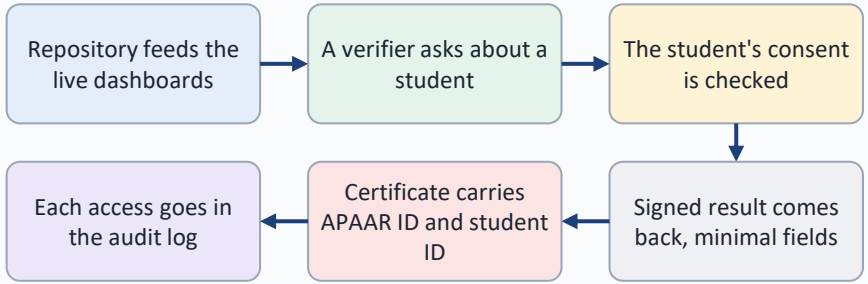
- A flag names the rule that raised it and gives the counts
- Supernumerary seats are added to approved intake before anything counts as excess
- A student found at two institutions is reported to both
- Audit entries are chained with SHA-256, so an edited row shows up

USER FLOW

Submission flow



Verification flow



A flag only asks an officer to look. The officer decides.

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TECHNICAL APPROACH

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INPUT

- CSV or Excel upload in bulk
- REST API for institution systems
- Intake, enrolment and supernumerary seats
- NRI, OCI and FN status, plus programme details

Proposed: Next.js upload, REST API

VALIDATION

- APAAR ID format and presence
- Duplicates, within one institution and across institutions
- Admissions above intake
- Fields that contradict each other

Proposed: rules, APAAR lookup adapter

REPOSITORY

- One record per APAAR ID
- Updates keep their history
- No APAAR ID yet: held as pending, not rejected
- Each change adds a hash-chained audit entry

Built: FastAPI, SQLAlchemy, audit chain

MONITORING

- Live dashboards for AICTE and institutions
- Intake use by programme
- Rule-based anomaly flags
- TF-IDF matching for near-duplicate names

Built: dashboards. Proposed: rules, TF-IDF

OUTPUT

- A verifiable record, with QR code and signed token
- Reports, with CSV export
- Verification API for authorised parties

Proposed: signed token, QR code

CONSENT AND PRIVACY

- The student's consent is recorded before anyone verifies
- A verifier receives only the fields it asked for
- Masking happens in the API, not just on screen

VERIFICATION AND API

- A verifier sends the APAAR ID and a purpose
- The answer is a signed result; certificates print the APAAR ID
- API and adapters connect AICTE and institution systems

SECURITY

- JWT login, bcrypt passwords
- Role checks on every endpoint
- Hash-chained audit log
- HTTPS in transit, encrypted storage

HUMANS DECIDE

- Nothing is rejected or penalised automatically
- A flag names the rule behind it
- Methods are named plainly: rules and TF-IDF

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PLATFORM, VALIDATION AND MONITORING

All approved institutions feed one repository keyed by APAAR ID; checks and monitoring sit on top.

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BUILT IN THE PROTOTYPE

- Login with JWT and role-based guards
- Audit log chained with SHA-256, plus a verify check
- Institution registry with an obligation and alert engine
- Compliance dashboard and notifications

PROPOSED FOR APAAR

- Student upload by file and API
- APAAR check through a registry adapter
- Intake-cap and cross-institution duplicate checks
- TF-IDF flag for near-duplicate names
- Consent-based verification with signed records
- Masking of student fields in the API

Institutions reporting	Records submitted	Valid	Flagged	Excess intake	Duplicate enrolment

DATA PER STUDENT

- APAAR ID (unique key)
- Programme, year, admission date
- Approved intake, actual enrolment
- Supernumerary seats
- NRI category, OCI status
- Foreign national (FN) status
- Consent flag and source

INTAKE CHECK (synthetic)

Flagged: 6 above allowed

Approved intake 120
Supernumerary seats 6
Allowed enrolment 126
Actual enrolment 132
An officer reviews the 6 extra records.
Nothing is rejected automatically.

Validation rules (a flag always says why)

Rule	Catches	Result
ID format	Missing, or not 12 digits	Fix and resubmit
Registry lookup	ID not found, or details differ	Held for review
Duplicate in file	Same ID twice in one upload	Flag
Across institutions	Same ID active at two institutions	Both alerted
Intake cap	Enrolment above intake plus supernumerary seats	Flag with count
Near duplicate	Similar name and date of birth, different ID	Suggestion only
Consistency	Fields that contradict, or mandatory ones missing	Flag

Institution dashboard (synthetic sample)

Institution	Intake	Enrolled	Flags	Status
Kaveri Valley College	120	132	3	Excess
Northfield Institute	60	58	1	Alert
Malabar Pharmacy College	40	40	0	Clear

CONSENT, PRIVACY, AUDIT

- The student's consent is recorded before a verifier sees anything
- A verifier gets only the fields it asked for
- Masking happens in the API, not just on screen
- Views and changes go to the audit chain

VERIFICATION

- Verifier sends an APAAR ID and a purpose
- Result: valid or not, with a signed token and QR
- Certificates print the APAAR ID beside the student ID
- Roles: AICTE, institution, verifier

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FEASIBILITY, IMPACT AND BENEFITS

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Existing situation vs this platform

Factor	Today (as the PS describes it)	Waymark APAAR Hub
Student data	PGDM: yearly detail; others: total enrolment only	Student-level records from every institution
Identity key	No shared student key	An APAAR ID on each record
Excess enrolment	Not visible beyond totals	Counted by programme, with the reason
Duplicate enrolment	Not visible across institutions	Flagged to both institutions
Verification	No central route	Consent-based, signed result

WHY IT IS FEASIBLE

- Open-source: Next.js, FastAPI, SQL
- Access, audit chain and alerts already built
- Rules and TF-IDF run without a GPU
- File upload needs no integration work

PRACTICAL IMPLEMENTATION

- APAAR lookup by adapter, once authorised
- REST API and file upload for institutions
- AICTE portal connected through adapters
- PostgreSQL, queue and storage at scale

Potential challenges	How we handle them
A student has no APAAR ID yet	Held as pending, not rejected
A lookalike is taken for a duplicate	A flag is a suggestion; an officer decides
Student data is sensitive	Consent, few fields, masking and audit
Older institution systems	CSV template and API both work

Impact and benefits

Feature	Impact	Benefit
APAAR-keyed records	One identity per student	Records that can be trusted and verified
Automated validation	Errors caught at upload	Cleaner data, less rework
Intake check	Excess enrolment shown with a count	Caught early
Cross-institution check	Duplicate enrolment alerts	One student, one seat
Live dashboards	AICTE and institutions see the same numbers	Monitoring in real time
Verification	With consent; signed result	Employers can rely on it
Reports	For policy, accreditation and regulation	Governance based on data
Audit and consent	Hash-chained log of each change	Access can be traced

Expected beneficiaries

AICTE	Institutions	Students	Employers	Regulators
authentic, current student data	one upload, clear errors	one identity, control through consent	quick, trusted checks	reports for policy

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RESEARCH AND REFERENCES

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1. APAAR Brochure, CIET-NCERT (Ministry of Education)
Relevance: APAAR as a 12-digit national academic ID.

2. Digital Personal Data Protection Act, 2023
Act No. 22 of 2023, Gazette of India, 11 August 2023. Relevance: consent and data principal rights.

3. Anomaly Detection: A Survey (2009)
Chandola, Banerjee, Kumar. ACM Computing Surveys 41(3), Article 15. Relevance: rule and statistical anomaly flags.

4. Lightweight String Similarity Approaches for Duplicate Detection (2025)
Journal of Informatics and Web Engineering, Vol. 4, Issue 3. Relevance: matching near-duplicate names.

5. National Education Policy 2020 (Ministry of Education)
Relevance: policy basis for the Academic Bank of Credits and a national student identity.

PROJECT LINKS

Source code
github.com/Karunesh-18/SIH-AICTE-APAAR

Prototype images
Open photo folder (Google Drive)

Demo video
Open video folder (Google Drive)

OFFICIAL REFERENCE

AICTE Approval Process Handbook 2024-2027
Relevance: approved intake and approval norms. The intake and seat figures in this deck are synthetic.

The prototype uses invented data and is not an official AICTE system.